

James Barry

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LinkedIn: James-J-Barry | GitHub: James-J-Barry | Portfolio: jbarry.tech

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, Rust, R, MATLAB

AI & Machine Learning: LLM Prompt Engineering, LLM Evaluation, MCP, Vector Embeddings, NLP, PyTorch, scikit-learn

Frameworks & Tools: React, Next.js, PostgreSQL, AWS, Docker, Splunk, Git, GitHub Actions (CI/CD), Linux, HPC (PBS/Slurm)

EDUCATION

University of Maryland

College Park, MD

B.S. Computer Science, Minor: Computational Finance GPA: 3.97/4.0

Expected Graduation May 2028

- **Coursework:** Machine Learning, Algorithms, Advanced Data Structures, Computer Systems, Data Science, Linear Algebra, Applied Probability & Statistics, Financial Markets & Financial Datasets
- **Leadership:** Logistics Director, Bitcamp Hackathon: led ~20 organizers for 1,400+ attendees; Education Chair, Kappa Theta Pi: ran professional development and corporate networking events for 70+ members

PROFESSIONAL EXPERIENCE

T. Rowe Price

Owings Mills, MD

Software Engineering Intern — AI Engineering & Application Development

May 2026 – Aug 2026

- Engineered a data-enrichment pipeline for an AI-powered AWS cost-optimization tool, surfacing **\$252,634/year** in savings across 517 EC2 instances via pricing lookups and automated recommendation logic in a React/TypeScript frontend.
- Optimized data-pipeline query architecture, expanding EC2 coverage from 515 to **4,154 instances** and uncovering an additional \$111,622/year in savings by resolving Splunk API result caps and data dilution in mixed-resource queries.
- Raised AI-generated report data quality from **76% to 96%** field coverage via a systematic audit of 79 fields across 15 panels.
- Diagnosed a critical failure by tracing a 58MB error response to a Splunk proxy misconfiguration, restoring access to **104,205 AWS resources** across 200 accounts by migrating 5 Python API modules.

Parsons Corp. @ ARL DoD Supercomputing Research Center

Aberdeen Proving Ground, MD

Software Engineer & Data Scientist Intern

Jun 2025 – Aug 2025

- Built and deployed a scalable evaluation system for semantic similarity algorithms over **100,000+ Army Test Incident Reports**, improving evaluator accuracy and efficiency.
- **Improved similarity search precision by 280%+** via multi-stage LLM summarization pipelines and prompt engineering.
- **Cut processing time by 8×** by parallelizing LLM analysis across 8-node HPC clusters using PBS batch jobs.

First Year Innovation and Research Experience (FIRE)

College Park, MD

Quantum Machine Learning Researcher

Aug 2024 – Dec 2025

- Engineered a Quantum PCA (QPCA) algorithm using Quantum Phase Estimation to extract eigenvalues without full covariance matrices, cutting theoretical complexity from $\mathcal{O}(N^3)$ to $\mathcal{O}(\text{poly}(\log N))$; simulated density matrix exponentiation for IonQ.
- Validated **86% reconstruction parity** with classical PCA on 16×16 MNIST digit images using Mean Squared Error.

PROJECTS

BarryBudget — Personal Finance Web & Mobile App (solo)

barrybudget.com

Next.js, React, TypeScript, React Native/Expo, PostgreSQL/Prisma, Plaid, Stripe, MCP

- Shipped a read-only MCP server (“Agent Access”) letting Claude Desktop, Claude Code, and other agents query a user’s budgets, transactions, and net worth via user-minted API tokens; it can never move money or edit budgets.
- Built the full stack solo (web, iOS/Android, backend): Plaid webhook-driven, cursor-based transaction sync with AES-256-GCM token encryption, passkey/TOTP two-factor auth, and usage-based Stripe billing with webhook-synced entitlements.
- Directed agentic development (Claude Code) via a living spec, with CI (Vitest, Playwright) gating every change.

Volatility Surface & Model-Divergence Engine

GitHub

Python, NumPy/SciPy, scikit-learn, Streamlit

- Engineered a live pipeline ingesting and validating 1,000+ real-time market records, feeding a Monte Carlo (GBM) simulation and a Newton-Raphson/Brent solver to recover latent model parameters.
- Cross-validated a constrained parametric fit against a gradient-boosted model of the live market surface, cutting model-vs-market error by **~90%** versus a naive baseline.